

THE STATES OF A MARKOV CHAIN

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STATE YOUR STATES CLEARLY TO REACH THE CORRECT CONCLUSION

Once a Markov chain has been modelled, a natural question to ask yourself is how its states behave over time. For example, can a given state always be reached? Will the chain eventually return to a particular state? Or can it become trapped in certain states forever? In order to answer these questions, we can classify the states of a Markov chain according to their long term behaviour. This leads to list of concepts: communication classes, irreducibility, recurrence, transience, periodicity, and absorption.

CLASSIFICATION OF STATES

State j is **accessible**, also known as **reachable**, from state i if there is a directed path in the transition graph from i to j , i.e., there exists n such that $p_n(i, j) > 0$. States i and j of a Markov chain **communicate** with each other, $i \leftrightarrow j$, if there exists $n, m \geq 0$ such that $p_n(i, j) > 0$ and $p_m(j, i) > 0$. That is two states communicate if each state has a positive probability of eventually being reached by a chain starting in the other state, in other words, each state is accessible from the other. The relation, $\leftrightarrow$, is an equivalence relation on the state space, i.e., it is reflexive, $i \leftrightarrow i$, since $p_0(i, i) = 1 > 0$; symmetric, $i \leftrightarrow j$ implies that $j \leftrightarrow i$; and transitive, $i \leftrightarrow j$ and $j \leftrightarrow k$ implies that $i \leftrightarrow k$. To see that transitivity holds, note that if $p_{n_1}(i, j), p_{n_2}(j, k) > 0$, then $p_{n_1+n_2}(i, k) = \mathbb{P}(X_{n_1+n_2} = k \mid X_0 = i) \geq \mathbb{P}(X_{n_1+n_2} = k, X_{n_1} = j \mid X_0 = i) = \mathbb{P}(X_{n_1} = j \mid X_0 = i) \mathbb{P}(X_{n_1+n_2} = k \mid X_{n_1} = j) = p_{n_1}(i, j) p_{n_2}(j, k) > 0$. Similarly, $p_{m_1}(j, i), p_{m_2}(k, j) > 0 \Rightarrow p_{m_1+m_2}(k, i) > 0$. This equivalence relation partitions the state space into disjoint sets called communication classes.

If there is only one communication class, i.e., $\forall i, j$ there exists an $n = n(i, j)$ with $p_n(i, j) > 0$, then the chain is **irreducible**. If for every state j that is accessible from i , i is also accessible from j , then states with this property are called **recurrent** states, and classes with this property are called recurrent classes, i.e., a Markov chain starting in a recurrent class never leaves that class. Other states/classes are called **transient**. Another way to describe recurrent and transient states is that a state i is recurrent if $\mathbb{P}(X_n = i \text{ infinitely often} \mid X_0 = i) = 1$ and transient if $\mathbb{P}(X_n = i \text{ infinitely often} \mid X_0 = i) = 0$. If state i is transient, the Markov chain will visit the state i for finitely many times and then leave i and never come back. Recurrence and transience are class properties, i.e., if C is a communicating class, then either all of its states are recurrent or all of its states are transient.

If it is impossible to leave a state, that is, $p(i, i) = 1, p(i, j) = 0 \forall j \neq i$, then the state is called **absorbing**. A Markov chain is absorbing if it has at least one absorbing state and if from every state it is possible to go to an absorbing state. This leads into equations for absorption probability, let a_i be the probability that the chain is eventually absorbed into a particular absorbing state, s . Thus, the probability to reach a specific absorbing state $s, a_1, \dots, a_M$, are unique solutions to equations $a_s = 1, a_i = 0 \forall$ absorbing states $i \neq s$, and

$a_i = \sum_{j=1}^M p(i, j) a_j \forall$ transient states i . These equations can be derived using the law of total probability by

conditioning the absorption probabilities on the next state. This leads nicely into the equations for working out the expected time to absorption. The expected times to absorption, $\mu_1, \mu_2, \dots, \mu_M$, are unique solutions to

the equations, $\mu_i = 0 \forall$ absorbing states i and $\mu_i = 1 + \sum_{j=1}^M p(i, j) \mu_j \forall$ transient states i . Again, these equations

can be derived using the law of total probability by conditioning the expected times to absorption on the next state. We add 1 since it takes one step to reach the next state.

Suppose that P is the matrix for an irreducible Markov chain, the period of a state $i, d = d(i)$, is the greatest common divisor of $J_i := \{n \geq 0 : p_n(i, i) > 0\}$. If P is reducible, we can consider separately each of the recurrent communication classes. A state is aperiodic if $d(i) = 1$. An irreducible matrix, P , is aperiodic if $d = 1$. A Markov chain is aperiodic if all of its states are aperiodic.

AN EXAMPLE: GAMBLER'S RUIN PROBLEM

A classic example of a Markov chain is the gambler's ruin problem. Consider a firm that currently has £1m of available liquidity. At the end of each trading day, market conditions may either improve or worsen the firm's liquidity position. Suppose that liquidity increases by £1m with probability $\frac{2}{3}$ and decreases by £1m with probability $\frac{1}{3}$. The firm will cease trading if liquidity falls to zero. However, the firm reckons they are financially secure once liquidity reaches £3m. What is the probability that the firm reaches financial security?

Solution: Let X_n denote the firm's liquidity level, in millions of pounds, after n trading days. The state space is $S = \{m\} = \{0, 1, 2, 3\}$, since the firm's liquidity can only take these values. The states of the Markov chain correspond to the firm's liquidity level, whereby, state 0 represents insolvency, whilst state 3 represents the target level. Thus, the transition matrix would be:

$$P = \{P_{ij}\} = \begin{bmatrix} p(0,0) & p(0,1) & p(0,2) & p(0,3) \\ p(1,0) & p(1,1) & p(1,2) & p(1,3) \\ p(2,0) & p(2,1) & p(2,2) & p(2,3) \\ p(3,0) & p(3,1) & p(3,2) & p(3,3) \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ \frac{1}{3} & 0 & \frac{2}{3} & 0 \\ 0 & \frac{1}{3} & 0 & \frac{2}{3} \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

The initial state is $X_0 = 1$, where the firm starts with £1m of available liquidity. At state 1, the next state is 0, where the firm decreases their liquidity position, with probability $\frac{1}{3}$, and 2 where the firm increases their liquidity position, with probability $\frac{2}{3}$. That is $p(1,0) = \frac{1}{3}$ and $p(1,2) = \frac{2}{3}$. Similarly, $p(2,1) = \frac{1}{3}$ and $p(2,3) = \frac{2}{3}$. The states 3, where the firm reaches financial security, and 0, where the firm ceases trading, are absorbing states. Let a_i denote the probability that the chain is eventually absorbed in state 3 given that it starts in state i . To calculate the probability that the firm reaches absorbing state 3, we can apply absorption probability

equations: $a_0 = 0, a_1 = \sum_{j=0}^3 p(1,j)a_j, a_2 = \sum_{j=0}^3 p(2,j)a_j, a_3 = 1$. Thus, using the transition matrix, we have $a_1 = \frac{1}{3} \times 0 + \frac{2}{3} \times a_2, a_2 = \frac{1}{3} \times a_1 + \frac{2}{3} \times 1 \Rightarrow a_1 = \frac{4}{7}, a_2 = \frac{6}{7}$. Thus, starting from £1m, the firm has $\frac{4}{7}$ probability of reaching the target level.

CONCLUSION

Classifying states in a Markov chain is incredibly useful for understanding long term behaviour. By analysing accessibility and communication, we can identify the structure of the state space and determine whether a chain is irreducible. Recurrence, transience, and periodicity help describe how frequently states are revisited, while absorbing states and probabilities allow us to analyse systems that will eventually terminate in a particular outcome. Examples of Markov chains are not just limited to the gambler's ruin problem, they can be found across a multitude of topics.

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